AIM261 Module Descriptor

AI Essentials: From Data to Agents

RYSERA STEM

Module Details

Course Code:	AIM261
Course Title:	AI Essentials: From Data to Agents
Hours/Week:	2 hours/week for 8 weeks (16 hours total)
Mode:	Online
Pre-requisites:	Basic Python (Free course provided)

Module Aim

This module provides an applied, engineering-focused introduction to Artificial Intelligence for post–A-Level students with some prior exposure to mathematics and programming. The course develops a strong foundation in data handling, machine learning algorithms, neural networks, computer vision, natural language processing, and modern agentic AI workflows. Students will gain practical, industry-relevant experience through hands-on experiments, real datasets, and a final AI project.

Learning Outcomes

Upon successful completion of the course, students will be able to:

- LO-1: Describe key concepts in modern AI, including data pipelines, learning systems, and intelligent agents.
- LO-2: Apply Python and scientific libraries to implement data processing and machine learning tasks.
- LO-3: Develop predictive models using supervised and unsupervised learning techniques.
- LO-4: Build and experiment with neural networks and computer vision pipelines.
- LO-5: Understand how large language models and transformer-based systems operate and apply them in workflows.
- LO-6: Construct simple agentic AI systems capable of reasoning and tool usage.

Module Outline

Week	Topics
Week 1	Introduction to AI: evolution of AI systems, narrow vs general AI, real-world applications, foundations of responsible and ethical AI
Week 2	Python essentials for AI: scientific computing with NumPy, data wrangling with Pandas, plotting with Matplotlib, efficient programming habits
Week 3	Data engineering: data collection, cleaning, preprocessing, handling missing values, scaling, encoding, dataset splitting; building a reusable data pipeline
Week 4	Machine learning fundamentals: supervised and unsupervised learning, decision trees, linear models, clustering, loss functions, evaluation metrics
Week 5	Neural networks and deep learning: perceptrons, activations, multilayer networks, gradient descent, backpropagation, introduction to deep learning workflows
Week 6	Image processing and computer vision: image representation, convolutional neural networks, transfer learning, hands-on image classification tasks
Week 7	Natural language processing and LLMs: tokenization, embeddings, transformers, prompt engineering, building simple chatbots
Week 8	Agentic AI: agent reasoning, decision-making, tool usage, workflow automation;